Model output report: ABOD_k60_pct

Summary of model_output_ABOD_k60_pct.csv (fusion of rule-based + ABOD_k60_pct scores).

1. Overview

Item	Value
Total customers	61,410
Flagged (top 5%)	3,071 (5.00%)
Not flagged	58,339
Effective threshold (risk_score ≥)	0.6498

Insight: This model flags the top 5% of customers by combined risk score (rule-based plus ABOD_k60_pct anomaly score), so 3,071 customers are marked for review. The threshold 0.6498 is the minimum score needed to be in that top 5%. All flagged customers should be prioritised for further investigation; the rest of the population is below the cutoff.

2. Risk score distribution

Statistic	All	Flagged	Not flagged
Min	0.0210	0.6498	0.0210
Max	0.9580	0.9580	0.6498
Mean	0.4681	0.6871	0.4566

Insight: The spread of scores (min to max, and standard deviation) shows how much the model separates customers. Flagged customers have higher mean and max scores than the rest; the gap between flagged and not-flagged means indicates how distinct the high-risk group is. IQR (interquartile range) is 0.150, so the middle 50% of scores span this range; a wider IQR suggests more variation in risk across the population.

3. Distribution by score band

Score band	Count	% of total
[0.0, 0.1)	446	0.7%
[0.1, 0.2)	439	0.7%
[0.2, 0.3)	3,165	5.2%
[0.3, 0.4)	12,655	20.6%
[0.4, 0.5)	18,801	30.6%
[0.5, 0.6)	18,995	30.9%
[0.6, 0.7)	6,147	10.0%

[0.7, 0.8)	719	1.2%
[0.8, 0.9)	42	0.1%
[0.9, 1.0)	1	0.0%

Insight: Most customers fall in the lower score bands; only a small share have scores in the 0.5+ range (42.2% here). The top 5% flagged set sits at the right tail of this distribution. If there are customers just below the threshold (within 0.02), there are 1,476 in that borderline band—these may warrant monitoring as small changes in behaviour could push them into the review set.

4. How to use this output

Use the corresponding model_output_*.csv and model_output_*_explanations.csv for Task 2 and Task 3. The explanations file gives a plain-English reason for each customer's score; the report above summarises the overall distribution and helps you understand how this model's fusion (rule + anomaly) behaves across the population.